Information and Inference: A Journal of the IMA (2026) **00**, 1–14
https://doi.org/

Beyond First Attendance: Interpreting Visit Frequency in University Mathematics Support

AUTHORS
Universidad de las Américas Puebla
*Corresponding author: mail@udlap.mx

[]

Recorded attendance at university mathematics-support centres may reflect academic help-seeking and incentives arising from other university programmes, complicating its interpretation as a dose of instruction. At Universidad de las Américas Puebla, first-semester students can complete one activity in a social and cultural integration programme (PPA1) by making three visits to the mathematics support centre (CMAT); this option was available throughout the 11 study periods. We examine 6,627 first eligible university mathematics-course attempts across 190 instructor–period contexts, with outcomes of standardised final performance (Z) and passing (PASS, numeric grade ≥ 7.5). Models adjust for instructor–period and degree programme; pairwise Wald comparisons use Holm multiplicity correction. Any recorded attendance was associated with 0.361 standard deviations higher adjusted Z (95% CI 0.302–0.421) and a 15.3-percentage-point higher adjusted PASS probability (95% CI 12.4–18.1). In the outcome-blind support-based user-only grouping, omnibus tests were not significant ($p = 0.317$ for Z, $p = 0.152$ for PASS), and no positive-frequency pair survived Holm adjustment. In a finer exploratory model including non-users, 7+ versus one visit was associated with a 14.5-percentage-point higher PASS probability (Holm-adjusted $p = 0.0206$), but the corresponding user-only comparison was not Holm-significant. The ranges 0, 1–3, 4–6 and 7+ are descriptive, not four confirmed outcome regimes. Our contribution is to distinguish initial attendance from the limited inferential resolution of repeated use, in a setting where a continuously available integration incentive may affect recorded visits. The analyses do not identify causal effects of either attendance or PPA1.

Keywords: mathematics support centre; mathematics learning support; visit frequency; Holm correction; multiple comparisons; academic performance; higher education

1. Introduction

University mathematics and statistics support centres provide supplementary learning opportunities for students entering programmes with different levels of prior mathematical preparation. Reviews have documented the development of such services and the methodological difficulty of evaluating them with attendance records alone (Matthews et al., 2013; Lawson et al., 2020; Mullen et al., 2024). Participation is a decision made by students within an academic and institutional context rather than an assigned intervention. A centre can be used to resolve an immediate question, repeatedly revisit difficult material, or satisfy a participation activity unrelated to mathematics coursework; consequently, equal visit counts need not represent equal instructional need or intensity. At this institution PPA1 is a first-semester social and cultural integration programme, in which activities can include attending sports events, theatre or dance performances or participating in student organisations. Completing three CMAT visits is one available PPA1 activity, offered throughout the study period, although actual credit receipt and individual eligibility are not observed for the complete MU cohort.

Prior research on university mathematics-support centres demonstrates that the difference between initial attendance and repeated participation has been an empirical concern for more than a decade. At

© The Author(s) 2026.

the National University of Ireland Maynooth, Mac an Bhaird et al. (2009) compared first-year students who attended the mathematics support centre more than once with those who attended once or never; among Arts and Finance students, the former group had an 89% pass rate compared with 62% for the latter, whereas the Science-student comparison was considerably weaker. Their use of Leaving Certificate mathematics achievement and diagnostic assessment also highlighted the importance of prior mathematical preparation, but their combined zero-or-one comparison does not resolve whether a first visit differs from non-attendance.

A more detailed distinction was made by Jacob and Ní Fhloinn (2019) using twelve years of administrative data from Dublin City University. Their logistic model controlled for previous mathematical achievement and module studied and reported passing odds ratios, relative to never attending, of 1.63 for a single visit and 13.78 for fifteen or more visits. These are associations in the *odds* of passing, rather than percentage-point effects or causal returns to additional visits. In a US Calculus I tutoring-centre study, Rickard and Mills (2018) used regression adjustment for prior ability and found that three tutoring visits were associated with approximately one percentage-point higher course grades. Importantly, results are not uniformly favourable: a smaller US drop-in tutoring pilot by Navarra-Madsen and Ingram (2010) found similar grades for mathematics majors who used the service and those who did not, with weak correlations between tutoring hours and grades. Together, these studies motivate distinguishing the service-use decision from the frequency profile rather than assuming a common graded benefit across institutions.

Evidence from Latin America provides an additional relevant comparison. At the Instituto Tecnológico de Costa Rica, Chacón-Vargas (2022) analysed participation in a first-year General Mathematics peer-tutoring programme with 528 enrolled participants in 2019. Although mean grades tended to be higher among students with more sessions, tests of grade distributions generally found no statistically significant differences except for a group attending at least fourteen of sixteen sessions; only 26 students belonged to that high-frequency group. This setting involves scheduled peer-tutoring and an additional preparatory programme rather than CMAT's drop-in faculty support, but it illustrates the interpretive challenges of frequency categories and small upper-tail groups in a closer regional context.

Differences between support centres are a further reason for caution when transporting observed associations. Byerley et al. (2024) assembled information from over 26,000 students at ten US mathematics tutoring centres, combining records of visits, prior preparation and grades with organizational characteristics to formulate hypotheses about centre structure. Their study did not formally establish the effectiveness of particular organizational features. The broader evaluation review by Mullen et al. (2024), which included 148 publications from twelve countries, likewise documents substantial differences in support formats, data sources and evaluation designs. Matthews et al. (2013) and Lawson et al. (2020) provide methodological and institutional perspectives on the development and evaluation of such services.

Attendance is behavioural rather than assigned. Surveys of students who do not engage with mathematics support document heterogeneous practical, academic and informational reasons for non-use (Mac an Bhaird et al., 2013). A postsecondary help-seeking meta-analysis finds that the association with achievement depends on the type of help sought (Fong et al., 2023), whereas a centre-entry count cannot distinguish those forms of engagement. Experimental encouragement studies also clarify the inferential boundary. Randomised messages increased take-up of peer tutoring without establishing a corresponding improvement in grades (Pugatch and Wilson, 2018); a randomised encouragement design for supplemental instruction similarly produced a clear first-stage participation response but imprecise academic-performance estimates (Paloyo et al., 2016). These interventions concern encouragement, not random assignment to the amount of mathematics tutoring received.

There is also a distinction between studying *why students attend* and studying their *final outcomes conditional on attendance*. Using drop-in provision at Coventry University, Gokhool and Lawson (2026) separately models whether students engage at all and the number of visits among those who engage; that study is about predictors of service use, not the effect of each visit on grades. Our data ask the complementary academic-outcome question. For positive attendance frequencies, testing every possible pair becomes difficult when the upper tail contains relatively few students and nominal p-values invite selective interpretation. The analysis therefore treats initial engagement and frequency among users as separate inferential objects, preserves an outcome-blind grouping decision and adjusts the full declared pairwise family using Holm's procedure.

This paper examines the association between recorded CMAT visit frequency and final outcomes in the first eligible attempt at a university mathematics course at Universidad de las Américas Puebla, Mexico. It separates the benchmark comparison of no recorded attendance versus any attendance from the central analysis among students who used the centre. We examine two outcomes: a within-instructor–period-standardised final-grade measure and the probability of passing. For an eligible first-semester student, a third CMAT visit can complete one PPA1 activity, while subsequent attendance goes beyond the requirement of that particular option. The visit count does not reveal whether any individual attended to obtain PPA1 credit. The primary inference uses instructor–period and degree-programme adjustment, cluster-robust omnibus tests and pairwise Wald contrasts whose multiplicity is controlled by the Holm procedure. The positive-frequency grouping originally selected from an outcome-blind support audit is reported separately from a more finely resolved exploratory attendance grid.

Our research questions ask whether adjusted outcomes differ between no use and any recorded use; whether positive frequencies can be distinguished after multiplicity correction; and how far the descriptive bands of zero, one to three, four to six and seven or more visits can be interpreted without claiming four empirically homogeneous groups. The $3|4$ boundary marks the first visit beyond a possible PPA1 activity for eligible students, not a proven academic threshold; $7+$ is an exploratory tail grouping. We assess sensitivity to grouping and instructor-level clustering. The study cannot identify a minimum effective visit count, measure the causal impact of PPA1 incentives or establish causal returns to attendance.

2. Institutional context

The Centro de Aprendizaje de Matemáticas (CMAT) is a mathematics support centre at Universidad de las Américas Puebla, Mexico. Its working space includes tables and whiteboards, and mathematics professors provide assistance outside regular classes. The existing institutional description records weekday availability from 08:00 to 17:50 during the study period. A student could receive help from someone other than their regular course instructor; a centre entry was recorded as a visit, without a corresponding measure of duration, tutoring content or individual instructional intensity.

The administrative attendance series covers 11 consecutive spring and autumn academic periods from spring 2019 to spring 2024. Throughout the study window, PPA1 operated as a first-semester programme of social and cultural activities intended to help new students integrate into university life. Students could participate by attending sporting events, theatre or dance performances, or engaging with student organisations; completing three CMAT visits was one possible PPA1 activity. This option remained available throughout the 11 observed periods, according to institutional information provided to the study. For an eligible participant, the third CMAT visit completes this particular PPA1 activity, whereas a fourth visit exceeds its requirement. However, the MU records do not establish PPA1

eligibility or actual credit receipt for every included student, nor do they reveal why an individual visited CMAT or which alternative PPA1 activities were selected. The programme's integration purpose is distinct from mathematical help-seeking, and the visit count may therefore mix academic and non-academic motives. Neither continuity of the programme nor the three-visit requirement constitutes an exogenous treatment threshold; a written copy of the institutional programme rules should be retained for documentation.

3. Data and methods

3.1. *Population, attendance and study window*

We analyse each student's first eligible observed attempt at Matemáticas Universitarias (MU) during an academic period covered by CMAT records. Administrative equivalence, revalidation and other transactions without an eligible instructor–period grading context are excluded. The resulting sample has 6,627 students across 190 instructor–period contexts.

Attendance is counted at the student–period level, including all centre visits within the same academic period as MU, irrespective of the subject label recorded for a visit. Zero visits means no *recorded* centre use, not absence of academic effort or need; a positive visit count measures entry frequency, not time spent receiving mathematical instruction.

3.2. *Frequency categories and the chronology of their selection*

The outcome-blind support audit originally selected exact 1, 2, 3, 4 and 5 visits and a pooled 6+ category for the principal comparison among users. This choice considered both student counts and overlap within the instructor–period groups required by the fixed-effect analysis. Distinguishing exact six from 7+ reduced the minimum instructor–period overlap between adjacent attendance groups from 26 to 15. Thus, the pooled-6+ categories constitute the more defensible pre-outcome support specification rather than a threshold chosen to maximise significant contrasts.

For descriptive transparency and an exploratory finer-grained comparison, we additionally report the zero-inclusive grid 0, exact 1–6 and 7+ visits. Counts for these eight levels are respectively 5,393, 517, 245, 170, 96, 55, 50 and 101. The finer top-code was adopted after the original support decision and is not presented as though it had been frozen before examining outcomes. We report adjusted comparisons both in the main among-user support-based analysis and in the zero-inclusive exploratory grid, with separately identified families of tests. For interpretation only, we overlay four institutional/descriptive bands: 0 (5,393 students), 1–3 (932), 4–6 (201), and 7+ (101). These bands highlight the PPA1 completion rule and the small upper tail, but they are not regrouped regression categories and no claim of four distinct adjusted effects follows from the display.

3.3. *Final academic outcomes*

The binary variable PASS equals one for a final numeric grade ≥ 7.5, including a grade of exactly 7.5, and equals zero for a numeric grade *strictly* below 7.5 or for the recorded administrative states baja voluntaria (BV), retiro temporal (RT) and baja académica (BA). PASS uses no numerical imputation for the administrative states. Descriptive outcome figures distinguish numeric failure ($Y < 7.5$) from administrative records without a numeric final grade (BV/RT/BA); the latter are not presumed to be voluntary decisions or equivalent academic situations.

The continuous outcome is standardised within the instructor–period grading context. Let Y_{ic} denote the completed numeric final value for student i in context c; then

$$Z_{ic} = \frac{Y_{ic} - \overline{Y}_c}{s_c}, \tag{1}$$

where $\overline{Y}_c$ and s_c are the mean and sample standard deviation of the completed grades within that instructor–period. Numeric grades are unchanged. For this continuous specification, BV, RT and BA are first assigned values below 7.5 using the existing context-specific imputation rule based on observed sub-passing numeric grades (kernel-density sampling with a fixed seed and appropriate fallback rules). Such assignments allow a single standardised final-outcome variable but introduce a modelling assumption that is absent from PASS. A sensitivity restricted to observed numeric final grades is therefore interpreted as a distinct complete-case population rather than the same outcome without uncertainty.

3.4. *Adjusted comparisons and multiplicity*

The zero-versus-any-attendance benchmark is estimated with instructor–period fixed effects, degree-programme indicators and standard errors clustered by instructor–period. We report adjusted standardised-grade differences in SD units and pass-probability contrasts from a fixed-effect linear probability model in percentage points.

The primary attendance-frequency question concerns students with one or more recorded visits ($n = 1,234$). Within this group, we fit categorical-frequency indicators alongside the same contextual and degree controls, first test the omnibus null of equal adjusted outcomes and then report pairwise Wald contrasts. The pooled-6+ grouping has six positive-frequency levels and therefore 15 pairs per outcome; the exploratory exact-six/7+ grouping has seven levels and 21 pairs per outcome. The separate zero-inclusive grid has eight levels and 28 pairs per outcome. The Holm procedure adjusts the resulting pairwise p-values *within each explicitly stated outcome and specification family*. Contrasts between adjacent frequencies are reported to assess local resolution; a wider comparison between one visit and the high-frequency tail is distinct from evidence for a successive-visit gradient.

The continuous model uses ordinary least squares and the binary model a linear probability specification. Wald confidence intervals and hypothesis tests use the cluster-robust covariance; a sensitivity reclusters by instructor to allow correlation across academic periods involving the same instructor. The zero-versus-1+ benchmark is treated as conceptually distinct from frequency contrasts within users.

Absence of statistical significance does not demonstrate equivalence of attendance categories, and we do not report formal equivalence tests because an educationally justified equivalence margin was not fixed independently of the observed estimates. The analysis is observational and no coefficient is interpreted as a causal return to additional CMAT visits.

4. Results

4.1. *Attendance distribution and unadjusted outcomes*

Of the 6,627 first eligible MU attempts, 5,393 students (81.4%) had no recorded CMAT attendance and 1,234 (18.6%) had at least one recorded visit. Figure 1 distinguishes three mutually exclusive final states for each visit frequency: PASS (numeric grade ≥ 7.5), numeric non-PASS (grade strictly < 7.5), and an administrative final state without a numeric grade (BV/RT/BA). The left panel reports percentages

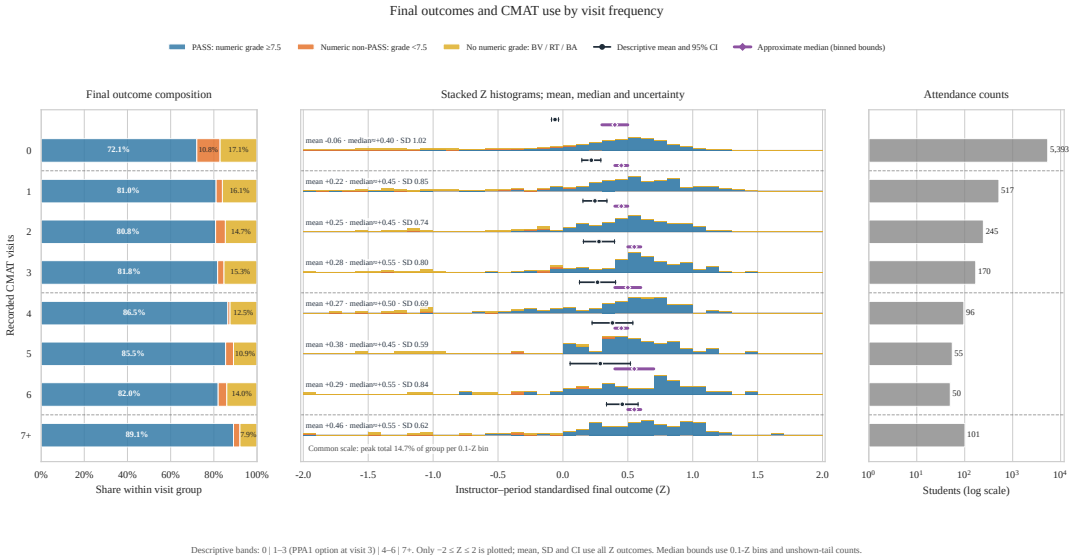


FIG. 1. Final outcome composition and recorded CMAT visits. Left: blue indicates PASS (observed numeric grade ≥ 7.5), orange indicates numeric failure ($Y < 7.5$), and gold indicates a non-numeric administrative final state (BV/RT/BA); all percentages use the full visit-group denominator. Centre: PASS, numeric non-PASS and BV/RT/BA are stacked (not overlapped) within each standardised Z bin; their heights sum to that bin's full-group frequency share; administrative outcomes enter only through the previously documented imputation rule for this continuous endpoint. Dark points with capped intervals show the *unadjusted* mean and its 95% confidence interval for each full attendance group, while the row text reports the group mean and standard deviation (SD). Purple diamonds and short bars display an *approximate* median and a conservative median-location interval derived from 0.1-Z bins and the number of observations outside the plotted window; these purple bars are not confidence intervals. The histogram itself is truncated to $-2 \leq Z \leq 2$, whereas the means, SDs and confidence intervals use the full outcome group. Right: numbers of students by visit frequency on a logarithmic axis. Horizontal separators display the descriptive attendance ranges 0, 1–3 (the PPA1 activity can be completed at visit three by eligible students), 4–6, and 7+, which are not statistically established regimes. The numeric PASS cutoff of 7.5 does not correspond to one universal Z-score threshold across instructor–period contexts.

of all students in each frequency, the centre stacks their mutually exclusive contributions within each instructor–period-standardised Z histogram bin, and the right panel shows group sizes. The displayed Z range is restricted to $[-2, 2]$, whereas the percentages and counts cover the full sample.

The distribution of visits is highly concentrated at zero, and cells beyond three visits contain substantially fewer students. The 1–3 range comprises 932 students, the 4–6 range 201, and the 7+ group 101. The within-frequency SDs of standardised Z range from 0.59 to 1.02; a group-specific SD need not equal one because standardisation was performed within instructor–period contexts, not within attendance groups. The displayed 95% intervals concern unadjusted group means rather than the adjusted between-frequency effects tested below. Median locations are only approximately identified from the available histogram bins and cannot be reported as exact sample medians without individual-level quantile summaries. Differences in the observed shapes and PASS fractions are descriptive; they do not establish a monotone benefit from additional visits.

Composition among non-passing students (exploratory). Among students without a passing outcome, 923 of 1,504 students with no recorded visits (61.4%) had an administrative rather than numeric final result, compared with 178 of 217 students with at least one visit (82.0%). The complementary

shares with observed numeric failure ($Y < 7.5$) were 38.6% and 18.0%, respectively. This comparison is *conditional on not passing*, so it must not be misread as a higher population-wide withdrawal rate among CMAT attendees or as evidence about the reasons for a BV, RT or BA record. These descriptive values use the pre-existing aggregates and do not change the prespecified Holm families of attendance-frequency contrasts.

4.2. *Benchmark: any use versus no recorded use*

After adjustment for instructor–period and degree programme, any recorded attendance was associated with 0.361 SD higher final standardised performance (95% CI 0.302–0.421, $p < 0.001$) and a 15.3 percentage-point higher PASS probability (95% CI 12.4–18.1 points, $p < 0.001$). This is the most precisely estimated separation in the study, but is an observational comparison of final outcomes and should not be construed as pre-to-post improvement from centre use.

4.3. *Primary frequency inference among CMAT users*

The originally selected positive-frequency grouping consists of exact 1, 2, 3, 4 and 5 visits and pooled 6+ visits, for a total of six levels. The fixed-effect omnibus test among the 1,234 users did not reject equal adjusted outcomes across these levels for continuous standardised performance ($p = 0.317$) or PASS ($p = 0.152$). Moreover, none of the 15 pairwise frequency contrasts per outcome remained significant after Holm adjustment.

The importance of prespecified multiplicity treatment is evident for the widest low-versus-high comparison. Between one visit and 6+ visits, the unadjusted pairwise p-values were 0.022 for the standardised outcome and 0.011 for PASS, but became 0.328 and 0.171 after Holm adjustment, respectively. Thus, isolated unadjusted comparisons provide insufficient evidence for differentiated positive-frequency categories under this primary specification.

4.4. *Eight visit levels and the exploratory four-range interpretation*

A descriptive overlay partitions the displayed frequencies into 0, 1–3, 4–6, and 7+. These ranges contain 5,393, 932, 201 and 101 students, respectively. The 1–3 range includes the three visits needed to complete one PPA1 social and cultural integration activity for eligible first-semester students; 4–6 describes continued use beyond that option, and 7+ isolates a sparse upper tail. Individual PPA1 eligibility and whether credit was awarded are not observed in these data. These four ranges motivate interpretation but have not been fitted or inferentially validated as four homogeneous regimes. The upper-triangular panels below retain all eight original visit categories and all 28 Holm-adjusted pairwise tests per outcome.

In the zero-inclusive grid $0, 1, 2, 3, 4, 5, 6, 7+$, the adjusted model of the continuous outcome rejected equality across all levels ($p < 0.001$). Every adjusted contrast between zero attendance and a positive visit category survived Holm adjustment across the 28 pairs; the weakest such comparison involved exactly six visits (adjusted $p = 0.042$). All seven zero-versus-positive PASS contrasts also survived Holm correction in the corresponding full-grid family.

A notable non-adjacent PASS comparison separated the 7+ group from one-visit users by 14.5 percentage points (95% Wald CI 5.9–23.1 percentage points, Holm-adjusted $p = 0.0206$). This estimate comes from the full sample of 6,627 and the 28-comparison family, not the primary user-only model; the finer split of the upper tail was made after the original support audit. The result therefore identifies an interesting exploratory high-frequency profile rather than a preselected attendance threshold.

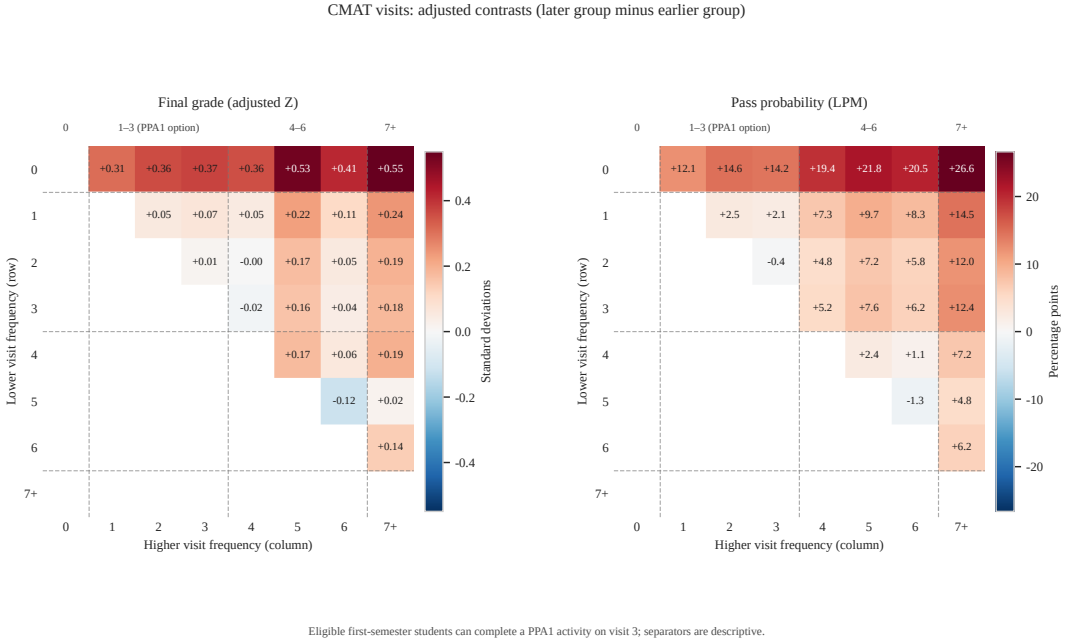


FIG. 2. Adjusted pairwise differences for instructor–period-standardised grades (SD units) and PASS probability (percentage points), using the eight-category reporting grid. Only the upper triangle is shown: positive values mean higher-frequency column minus lower-frequency row; shading encodes adjusted effect magnitude and direction, not statistical significance. Lines delimit hypothesised 0|1–3|4–6|7+ ranges, not Holm-established changepoints.

Within the seven positive-frequency categories, separate user-only omnibus tests remained inconclusive ($p = 0.213$ for standardised grades and $p = 0.102$ for PASS), and no positive-frequency pair survived the corresponding Holm adjustment. These findings are not numerically interchangeable with the full zero-inclusive model because removing zero-attendance students changes the sample, nuisance fit and covariance estimates in addition to the comparison family.

No adjusted adjacent visit contrast among users survived multiplicity correction in the reported analyses. The 3-versus-4 comparison does not establish an academic-outcome jump beyond the third visit, nor does the 2-versus-3 comparison establish a jump at possible completion of a PPA1 activity. Neither contrast estimates whether the social and cultural integration incentive caused attendance, because eligibility and actual participation credit are not observed for each MU student. Descriptively the mean Z changed from 0.267 to 0.382 between four and five visits and from 0.288 to 0.457 between exact six and 7+, but the corresponding adjacent adjusted contrasts did not establish a break.

4.5. *Sensitivity to upper-tail grouping and level of clustering*

The pooled-6+ primary specification remained inconclusive among users after Holm correction, whereas the full eight-group model identified the exploratory 7+ versus one-visit PASS comparison described above. This difference should be retained in interpretation rather than presented as if a uniquely optimal upper-tail resolution had been established.

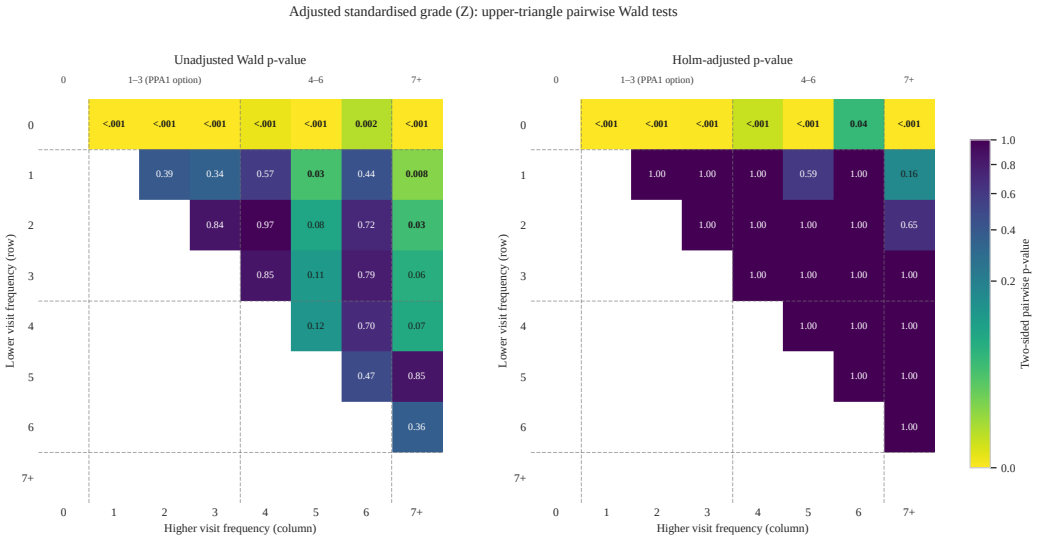


FIG. 3. Raw and Holm-adjusted pairwise Wald *p*-values for the standardised final-outcome model, full eight-category family (upper triangle only).

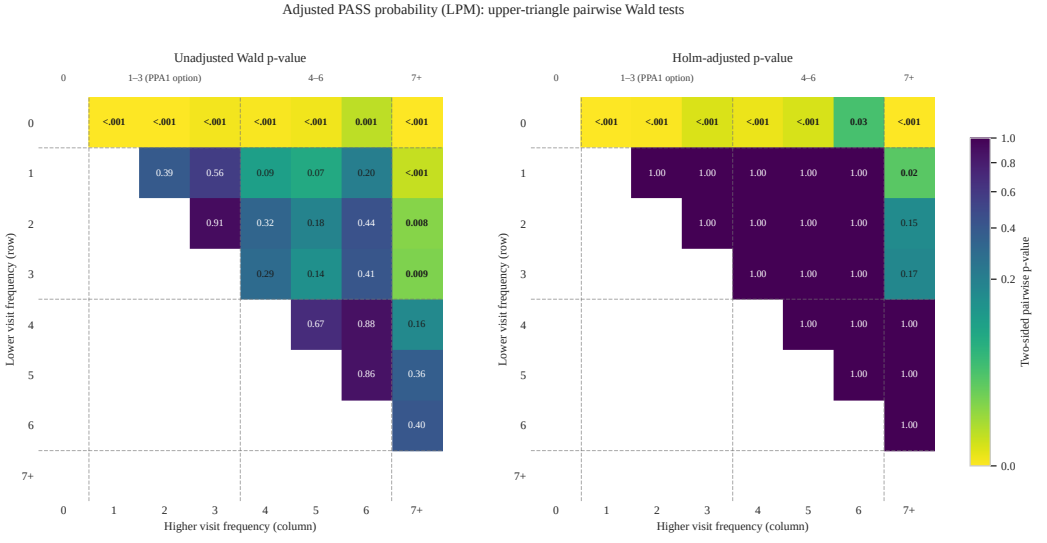


FIG. 4. Raw and Holm-adjusted pairwise Wald *p*-values for PASS under the linear-probability model, full eight-category family (upper triangle only).

Because the same instructor can teach in more than one period, we re-estimated uncertainty by clustering at instructor level. The any-use benchmark remained 0.361 SD (95% CI 0.305–0.417) and 15.3 PASS percentage points (95% CI 12.8–17.8). In the user-only pooled-6+ model, the instructor-cluster omnibus test yielded $p = 0.0477$ for grades and $p = 0.0551$ for PASS. None of the grade contrasts survived Holm correction, but the non-adjacent PASS contrast comparing 6+ visits with one visit was +12.0 percentage points (95% CI 4.3–19.8; adjusted $p = 0.0358$). No adjacent PASS contrast was significant. This sensitivity allows a wider low-versus-high distinction in PASS, but does not support a monotonic visit-by-visit gradient.

4.6. *Observed-numeric outcome check*

The complete-case analysis that excludes BV, RT and BA changes the analytic population and addresses a different endpoint. Across positive visit frequencies its omnibus test for the numeric-only standardised grade did not reject equality ($p = 0.397$). This check does not remove concerns about selection into an observed final numeric grade and cannot establish that the imputation rule is consequence-free; it supports reporting the distinct PASS outcome alongside the standardised final-outcome measure.

5. Discussion

The distinction between initial attendance and repeated attendance is central to interpreting routine mathematics-support records. The strongest result is the higher adjusted final performance and PASS probability of students with at least one recorded CMAT visit compared with those with none. This broad separation persists after controlling for instructor–period and degree programme, although the analysis is observational and cannot identify what would have happened if a non-attendee had been assigned to visit the centre.

The more demanding question is whether there is a reliable progression in outcomes from one positive visit frequency to the next. Under the positive-frequency grouping selected from outcome-blind support, the omnibus tests and every Holm-adjusted pairwise comparison were inconclusive. This does not establish equality or prove that additional support cannot matter; the high-frequency cells are sparse, the available information does not include pre-visit skill or tutoring duration, and the correction explicitly protects against selecting the most favourable result from many comparisons. Most importantly, the absence of multiplicity-supported *adjacent* contrasts is inconsistent with an empirically established staircase of improvements after successive visits.

There are nevertheless reasons not to conclude that all positive frequencies have identical associations. In the more detailed, zero-inclusive model, the seven-or-more-visits category had a higher adjusted PASS probability than the one-visit category after Holm correction; with instructor-level clustering in the originally grouped user-only model, the corresponding broader 6+ versus one-visit PASS difference was also significant. The different conclusions arise from specifications with different group support, samples, and covariance assumptions, and the fine-grained upper tail was not the grouping originally frozen before outcome inspection. The appropriate interpretation is therefore that *broader low-versus-high differences remain plausible*, whereas the available contrasts do not justify selecting a sequence of precise attendance changepoints.

The PPA1 programme's institutional purpose matters for interpreting recorded CMAT attendance. PPA1 offers first-semester students opportunities for social and cultural integration, including sports events, performing arts and student organisations; completing three CMAT visits is one alternative qualifying activity. This option operated throughout the eleven periods studied. Eligible students may

therefore visit for reasons not solely related to mathematical difficulties, while four or more visits go beyond the number required by this particular option. However, actual PPA1 eligibility, credit received, motivation and choices of alternative activities are not observed for every member of the MU cohort. The four descriptive ranges, 0, 1–3, 4–6 and 7+, are an interpretive framework rather than four inferred homogeneous groups. Neither the 2|3 nor the 3|4 comparison establishes an incentive effect or a causal threshold in academic performance.

The descriptive composition of non-passing outcomes suggests an additional research question. Students who attended CMAT and nevertheless did not pass had a greater proportion of BV/RT/BA records instead of numeric grades below 7.5 than non-attendees who did not pass. A previously fitted ancillary instructor–period-adjusted comparison of administrative versus numeric non-PASS outcomes estimated a 13.2-percentage-point difference (95% CI 6.8–19.6; nominal $p < 0.001$). This analysis is *exploratory*, conditional on non-PASS and outside the article's prespecified Holm families. Scholarship retention offers one possible explanation because institutional scholarships can require a minimum cumulative average; however, withdrawal limits also differ by scholarship scheme, and the available records do not identify scholarship recipients, reasons for withdrawing, or whether an individual BV, RT or BA actually protected academic standing. It would be incorrect to label all administrative outcomes as deliberate grade avoidance or as evidence that students exhausted academic effort before withdrawing.

The contribution is educational and evaluative rather than a new statistical procedure. Prior studies have already related mathematics-support frequency to academic outcomes (Jacob and Ní Fhloinn, 2019; Rickard and Mills, 2018); accordingly, the novelty is not the visit counter or Holm correction considered separately. Our analysis distinguishes the relatively clear first-attendance association from the limited statistical resolution among positive frequencies, in a setting where an integration-programme activity could also contribute to attendance. Instructor–period adjustment, outcome-blind support criteria for the primary grouping and explicit Holm families constrain the interpretation of unadjusted trends as dose responses or attendance thresholds. The implication for other universities is that centre-entry logs can be an imperfect proxy for demand for mathematics instruction, particularly where participation incentives coexist with academic support, and cannot by themselves justify an optimal number of mandatory visits.

6. Directions for future research

Two institutional mechanisms deserve dedicated evaluation beyond the present observational Holm comparison. First, the relationship between mathematics-support attendance, scholarship eligibility and the administrative resolution of a non-passing course should be studied using authorised, de-identified information on scholarship type, prior cumulative average, the applicable historical policy on failed grades and withdrawal counts, timing of CMAT visits and withdrawals, and, where ethically and practically feasible, direct evidence of students' reasons for their decisions. A low numeric grade may threaten a minimum average, but an administrative withdrawal may itself count against scholarship conditions; therefore grade-protection behaviour remains a hypothesis to test, not a finding of this paper.

Second, the PPA1 social and cultural integration programme's continuously available three-visit CMAT option motivates an evaluation of whether that activity increases initial attendance, completion of the three visits, continued mathematics-support use or academic outcomes. Such research would require observed first-semester eligibility, actual PPA1 credit, alternative activities available and chosen, visit dates and a defensible comparison group or incentive experiment. Since the option was continuously available, the present sample has no identified introduction or withdrawal of the incentive

to exploit as a policy change. Comparisons between visits two and three or three and four do not establish its causal impact.

7. Limitations

The primary concern is self-selection. Students are not randomly assigned to CMAT or to a particular visit count; academic need, existing mathematical preparation, motivation, schedules and instructor encouragement may influence both attendance and final performance. Adjustment for instructor–period and degree programme does not account for every relevant baseline characteristic. A sufficiently complete, comparable university entrance examination measure would permit an additional observed-preparation sensitivity but not, by itself, a causal interpretation.

Attendance and final outcomes are recorded within the same period. Students may visit because they encounter difficulties, return when support seems useful, or stop accumulating visits because they withdraw before the end of the period. The attendance records do not document duration or instructional content, so visit count cannot be equated with the delivered quantity of mathematical instruction. The continuous availability of the PPA1 CMAT activity during the study period is supplied by the institutional project context, but its application to individual students, actual credit awarded and reasons for attendance remain unobserved. Archiving the written activity guidelines would strengthen documentary traceability.

Upper-frequency cells are small, especially exact six visits and the pooled 7+ tail, reducing within-context overlap and precision. The pooled-6+ specification was selected with a support audit before reviewing outcomes, but the more detailed 7+ contrast was examined subsequently; the distinction limits its confirmatory interpretation. The observed instructor-level clustering sensitivity also shows that inferential conclusions for widely separated visit groups can depend on how repeated instructor contributions are treated.

Finally, the standardised continuous outcome relies on a documented numerical representation of administrative results, whereas PASS does not, and the complete-case numeric sensitivity conditions on receiving a final numeric grade. The two main outcomes consequently have distinct interpretation and measurement assumptions. Holm controls family-wise false rejections within the declared pairwise family when its tests are calibrated, but does not correct unmeasured confounding, specification choices or broader research-wide exploration; non-rejection is not proof of equivalence.

8. Ethics, data governance, and reproducibility

The study uses institutional administrative data. Protected student-level records will not be publicly released, and any shared outputs must be disclosure-reviewed aggregates or synthetic examples. Grouping choices, analytical code and reproducible figure-building procedures are maintained under version control.

9. Conclusion

Among 6,627 first eligible university mathematics-course attempts, the coexistence of academic help-seeking and an available three-visit social-and-cultural integration activity makes the distinction between 1–3 and 4+ institutionally meaningful, but individual PPA1 eligibility is unknown and the existence of four separate academic-outcome regimes remains unproven. Recorded CMAT use was associated with higher final standardised grades and a greater adjusted probability of passing than zero

use. However, the originally support-based comparisons among positive attendance frequencies yielded no Holm-significant pairwise contrasts, and no successive-visit gradient was reliably resolved. A higher PASS probability for very frequent users emerged in exploratory full-grid or alternative clustering specifications, but its sensitivity to grouping and inferential population prevents treating a particular visit count as an established academic threshold. The results support distinguishing initial engagement from frequency among attendees when evaluating mathematics-support services, while leaving causal effects of attendance undetermined.

Acknowledgments

Funding: None declared.

Generative AI tools were used to assist with code development, manuscript drafting, editing and language refinement; responsibility for the analysis, interpretation and final text remains with the authors.

References

C. Byerley, C. Johns, D. Moore-Russo, B. Rickard, C. James, M. Mills, B. Mammo, J. Oien, L. Burks, W. Heasom, M. Ferreira, C. Farthing, and D. Moritz. Towards research-based organizational structures in mathematics tutoring centres. *Teaching Mathematics and its Applications*, 43(1), 2024. doi: 10.1093/teamat/hrac026.

E. Chacón-Vargas. La tutoría entre pares y su incidencia sobre el rendimiento académico en un curso de matemática universitaria. *Revista Electrónica Educare*, 26(1):1–18, 2022. doi: 10.15359/ree.26-1.20.

C. J. Fong, C. Gonzales, C. Hill-Troglin Cox, and H. B. Shinn. Academic help-seeking and achievement of postsecondary students: A meta-analytic investigation. *Journal of Educational Psychology*, 115(1):1–21, 2023. doi: 10.1037/edu0000725.

F. Gokhool and D. Lawson. Demographic predictors of engagement with mathematics and statistics support. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 45(3):434–453, 2026. doi: 10.1093/teamat/hrag003.

M. Jacob and E. Ní Fhloinn. A quantitative, longitudinal analysis of the impact of mathematics support in an irish university. *Teaching Mathematics and its Applications*, 38(4):216–229, 2019. doi: 10.1093/teamat/hry012.

D. Lawson, M. Grove, and T. Croft. The evolution of mathematics support: a literature review. *International Journal of Mathematical Education in Science and Technology*, 51(8):1224–1254, 2020. doi: 10.1080/0020739X.2019.1662120.

C. Mac an Bhaird, T. Morgan, and A. O'Shea. The impact of the mathematics support centre on the grades of first year students at the national university of ireland maynooth. *Teaching Mathematics and its Applications*, 28(3):117–122, 2009. doi: 10.1093/teamat/hrp014.

C. Mac an Bhaird, O. Fitzmaurice, E. Ní Fhloinn, and C. O'Sullivan. Student non-engagement with mathematics learning supports. *Teaching Mathematics and its Applications*, 32:191–205, 2013. doi: 10.1093/teamat/hrt018.

J. Matthews, T. Croft, D. Lawson, and D. Waller. Evaluation of mathematics support centres: a literature review. *Teaching Mathematics and its Applications*, 32(4):173–190, 2013. doi: 10.1093/teamat/hrt013.

C. Mullen, E. Howard, and A. Cronin. A scoping literature review of the impact and evaluation of mathematics and statistics support in higher education. *Educational Studies in Mathematics*, 117:1–22, 2024. doi: 10.1007/s10649-024-10332-6.

J. Navarra-Madsen and P. Ingram. Mathematics tutoring and student success. *Procedia - Social and Behavioral Sciences*, 8:207–212, 2010. doi: 10.1016/j.sbspro.2010.12.028.

A. R. Paloyo, S. Rogan, and P. Siminski. The effect of supplemental instruction on academic performance: An encouragement design experiment. *Economics of Education Review*, 55:57–69, 2016. doi: 10.1016/j.econedurev.2016.08.005.

T. Pugatch and N. Wilson. Nudging study habits: A field experiment on peer tutoring in higher education. *Economics of Education Review*, 62:151–161, 2018. doi: 10.1016/j.econedurev.2017.11.003.

B. Rickard and M. Mills. The effect of attending tutoring on course grades in calculus i. *International Journal of Mathematical Education in Science and Technology*, 49(3):341–354, 2018. doi: 10.1080/0020739X.2017.1367043.